

The Geometry of Casting

- 1 Castable Objects
 - manufacturing with molds
 - polyhedra and castable objects
- 2 Half Plane Intersection
 - problem statement
 - a divide and conquer algorithm
- 3 A Plane Sweep Algorithm
 - representation of the output
 - merge edges
 - correctness and cost analysis

MCS 481 Lecture 10
Computational Geometry

Jan Verschelde, 16 September 2026

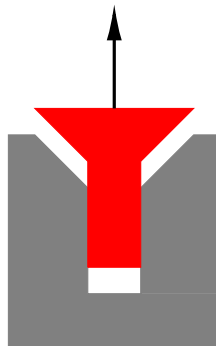
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manufacturing with molds

Metal objects are often manufactured by *casting*:

- 1 liquid metal is poured into a mold,
- 2 the liquid solidifies,
- 3 the object is removed from the mold.

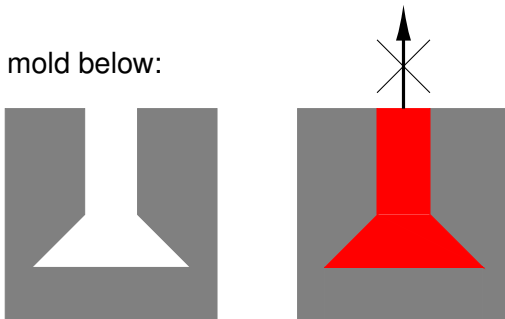


Two constraints:

- 1 the top of the object must have a flat surface,
- 2 we do not want to break the mold when removing the object.

Do not break the mold!

Consider the mold below:



Removing the solidified liquid is not possible without breaking the mold.

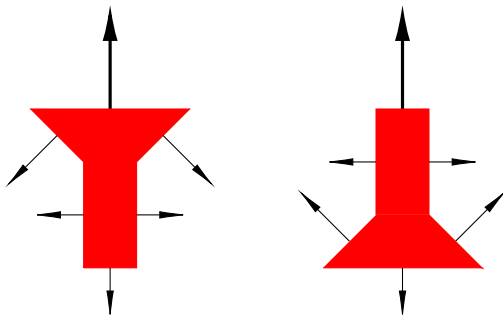
comparing the orientation of the objects

Compare the orientation, relative to the direction of removal:



outward normals to the facets of the objects

Consider outward pointing normals, perpendicular to the facets:



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polyhedra and outward normals

Definition (polyhedra and polytopes)

A **polyhedron** is the intersection of a finite number n of half planes

$$a_i x + b_i y + c_i z \leq d_i, \quad i = 1, 2, \dots, n.$$

A **polytope** is a bounded polyhedron.

Example: the unit cube is defined by 6 inequalities:

$$-x \leq 0, \quad x \leq 1, \quad -y \leq 0, \quad y \leq 1, \quad -z \leq 0, \quad z \leq 1.$$

The vertices of the unit cube are

$(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0), (1, 0, 1), (0, 1, 1), (1, 1, 1).$

Definition (outward normals and facets)

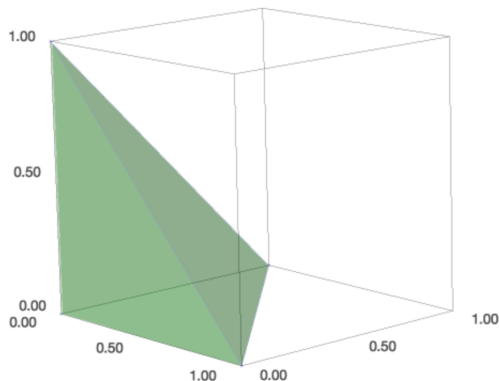
Every half plane $ax + by + cz \leq d$ has an **outward normal** (a, b, c) .

A 2-dimensional face of a polyhedron is called a **facet**.

A 1-dimensional face of a polyhedron is called an **edge**.

A 0-dimensional face of a polyhedron is called a **vertex**.

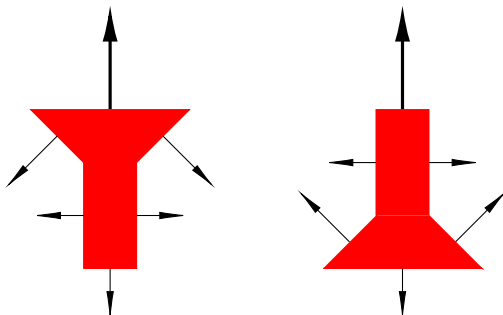
the tetrahedron



Exercise 1: The tetrahedron is the 3-dimensional version of the triangle. It is the polytope spanned by all convex combinations of the points $\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

How many half planes are needed to define the tetrahedron?
Compute the coefficients (a_i, b_i, c_i, d_i) of the half planes which intersection defines the tetrahedron (*see the script and notebook*).

necessary and sufficient conditions



Lemma (the removable criterion)

A polyhedron P can be removed by a horizontal translation if and only if all outward pointing normals to the nonhorizontal facets of P make an angle of at least $\pi/2$ with the vector $(0, 0, 1)$.

proof by contraposition

Lemma (the removable criterion)

A polyhedron P can be removed by a horizontal translation if and only if all outward pointing normals to the nonhorizontal facets of P make an angle of at least $\pi/2$ with the vector $(0, 0, 1)$.

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

$\Rightarrow$: By contraposition: suppose there is one outward normal with angle strictly less than $\pi/2$, then we cannot remove P .

$\Leftarrow$ follows on next slide

proof by contraposition, the other direction

Lemma (the removable criterion)

A polyhedron P can be removed by a horizontal translation if and only if all outward pointing normals to the nonhorizontal facets of P make an angle of at least $\pi/2$ with the vector $(0, 0, 1)$.

Proof. If and only if means $\Rightarrow$ and $\Leftarrow$.

$\Leftarrow$: By contraposition: suppose we cannot remove P , then we must show there is at least one outward normal with angle $< \pi/2$.

Let $q \in P$ be the point where q collides with facet f of the mold.

Then q is about to move into the mold, which implies that the outer normal to f makes an angle greater than $\pi/2$ with $(0, 0, 1)$.

But then $(0, 0, 1)$ makes an angle $< \pi/2$ with the outer normal to f .

Q.E.D.

castable objects

In order to test whether a polyhedron P is castable, we try all facets as top facets.

For every candidate top facet, we compute the angles with all other $n - 1$ facets. If all angles are at least $\pi/2$, then the object is castable.

Theorem (castable objects criterion)

Let P be a polyhedron with n facets.

We can decide whether P is castable in $O(n^2)$ time.

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half plane intersection

The input to our problem is a set of n triplets:

Let $H = \{ H_1, H_2, \dots, H_n \}$ be a set of half planes,

$$H_i = \{ (x, y) \mid a_i x + b_i y \leq c_i \}, i = 1, 2, \dots, n.$$

Output: $\bigcap_{i=1}^n H_i$ intersection of all half planes.

Observe the following:

- a half plane is convex,
- the intersection of any two convex sets is convex,

thus the intersection of n half planes is convex.

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a divide and conquer algorithm

Algorithm INTERSECTHALFPLANES

Input: $H_i = (a_i, b_i, c_i)$, $i = 1, 2, \dots, n$.

Output: $C = \bigcap_{i=1}^n H_i$, $H_i = \{ (x, y) \mid a_i x + b_i y \leq c_i \}$.

- 1 if $n = 1$ then
- 2 $C = H_1$
- else
- 3 split H into H_1 and H_2 , $\#H_1 = \lceil n/2 \rceil$, $\#H_2 = \lfloor n/2 \rfloor$
- 4 $C_1 = \text{INTERSECTHALFPLANES}(H_1)$
- 5 $C_2 = \text{INTERSECTHALFPLANES}(H_2)$
- 6 $C = \text{INTERSECTCONVEXREGIONS}(C_1, C_2)$

applying the algorithm MAPOVERLAY of Lecture 6

to compute $C = \text{INTERSECTCONVEXREGIONS}(C_1, C_2)$

The algorithm MAPOVERLAY takes $O((n + k) \log(n))$ time

- on inputs of size n , and
- on outputs of size k .

What is k for this problem?

The polygonal region C can have at most n half planes, therefore, at most n vertices and at most n edges, so $k \leq n$.

Thus, step 6 of INTERSECTHALFPLANES takes $O(n \log(n))$ time. The algorithm INTERSECTHALFPLANES has thus time $T(n)$:

$$T(n) = \begin{cases} O(1) & \text{if } n = 1, \\ O(n \log(n)) + 2T(n/2) & \text{if } n > 1. \end{cases}$$

Resolving the recurrence for $T(n)$ yields $O(n(\log(n))^2)$.

a familiar recurrence

Exercise 2: Show that

$$T(n) = \begin{cases} O(1) & \text{if } n = 1, \\ O(n \log(n)) + 2T(n/2) & \text{if } n > 1. \end{cases}$$

becomes $O(n \log(n))$ if the $O(n \log(n))$ in the formula for $T(n)$ is replaced by $O(n)$.

We can do better than applying MAPOVERLAY of Lecture 6 to compute $C = \text{INTERSECTCONVEXREGIONS}(C_1, C_2)$.

intersections in CGAL

Following

`AABB_polyhedral_facet_intersection_example.py`
from the `cgal-swig-bindings`:

- 1 Construct a tetrahedron, enumerate vertices, edges, facets.
- 2 Compute a segment query.
- 3 Compute a plane query.

The AABB tree component offers a static data structure and algorithms to perform efficient intersection and distance queries against sets of finite 3D geometric objects.

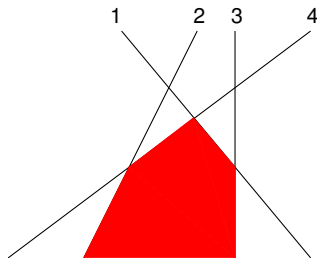
See the posted Jupyter notebook.

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representation of the output

How to store $\bigcap_{i=1}^n H_i$? Consider for example:

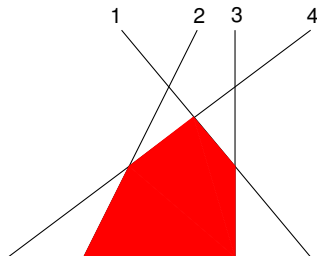


The top vertex $\text{top}(C)$ separates

- $\text{left}(C) = [H_4, H_2]$ from
- $\text{right}(C) = [H_1, H_3]$.

Left and right boundaries are stored by sorted half planes.

special cases



- What if there is no top vertex $\text{top}(C)$? Take ∞ .
- What if the top is horizontal? Take leftmost as $\text{top}(C)$.

DCEL of sorted half planes

Representing a convex region via the intersection of half planes:

- The region C defined by the intersection of half planes is stored as a sorted list of half planes, sorted from top to bottom.
- Vertices are computed by intersecting consecutive boundary lines.

Exercise 3:

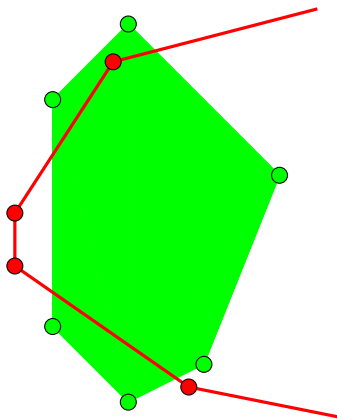
- Write pseudo code to construct a doubly connected edge list of a convex region given by the sorted half planes of the left and right boundaries.

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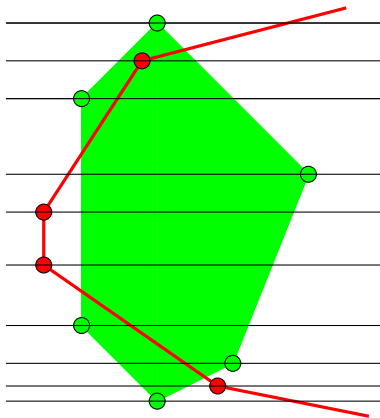
merge edges

Intersect the green polygon with the right of the right edges:



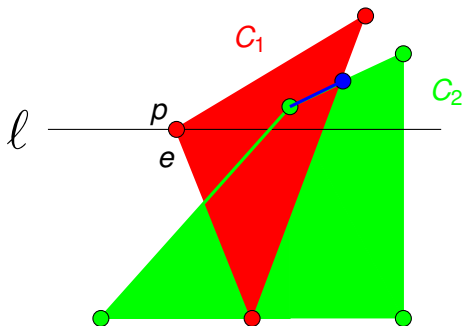
moving a sweep line

The event points are the end points of the edges:



handling event points

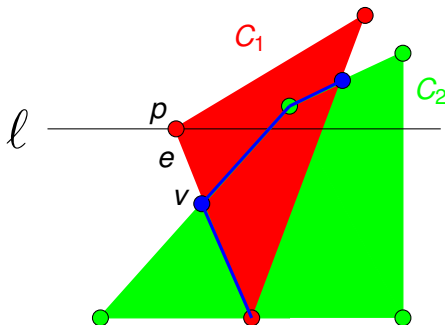
Consider the event point p , the upper end point of edge e , e lies on a half plane of $\text{left}(C_1)$:



Does e contribute to $C = C_1 \cap C_2$?
Yes, because e intersects $\text{left}(C_2)$.

the order of adding half planes

Consider the event point p , the upper end point of edge e , e lies on a half plane of left(C_1):

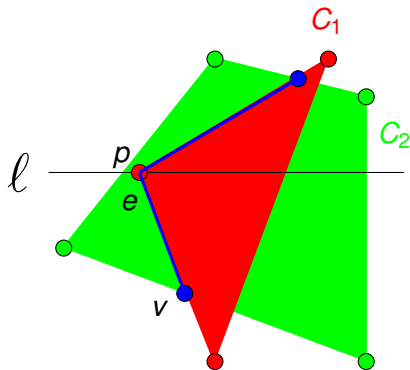
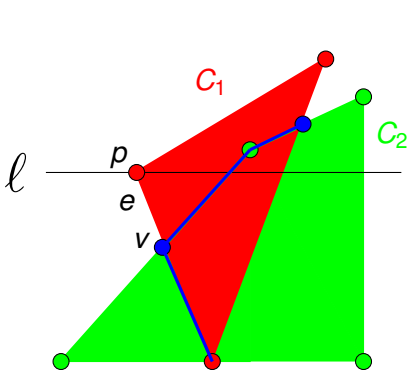


The half plane through e should be added only after the intersection point v is detected.

two cases

At an event point:

- Sometimes we add more than one half plane.
- The order of adding matters.



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correctness

For the correctness of the algorithm INTERSECTHALFPLANES to compute C , we need to show that

- 1 all half planes bounding the edges of C are added, and
- 2 the half planes are added in the correct order.

Exercise 4:

- 1 Describe the algorithm INTERSECTCONVEXREGIONS with pseudo code.
- 2 Show that the order of the tests implies that the half planes are added in the correct order.

cost analysis

Theorem (cost of convex region intersection)

The cost of INTERSECTCONVEXREGIONS is $O(n)$, for n half planes.

Then the time $T(n)$ for INTERSECTHALFPLANES has the recurrence

$$T(n) = \begin{cases} O(1) & \text{if } n = 1, \\ O(n) + 2T(n/2) & \text{if } n > 1. \end{cases}$$

This recurrence has $O(n \log(n))$ as solution.

Corollary (cost of half plane intersection)

The intersection of n half planes can be computed in $O(n \log(n))$ time and $O(n)$ storage.

summary and exercises

We covered sections 4.1 and 4.2 in the textbook.

- A problem from manufacturing industry provided motivation.
- We defined a plane sweep algorithm to compute the intersection of n half planes with a $O(n \log(n))$ running time.
- Half plane intersection is related to the convex hull problem.

Consider the following activities, listed below.

- 1 Explore the relevant code in CGAL.
- 2 Write the solutions to exercises 1, 2, 3, and 4.
- 3 Consider the exercises 2, 4, 5 in the textbook.